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Q12.125

$$\begin{cases} W_B = 5.443 \text{ N}, & m_B = 0.555 \text{ kg} \\ W_A = 13.667 \text{ N}, & m_A = 1.388 \text{ kg} \end{cases}$$

FBD

$$\textcircled{A} \quad \begin{cases} \Sigma F_y = m_A g + N_B - N \cos 30^\circ = m_A a_A \sin 30^\circ \quad \textcircled{1} \\ \Sigma F_x = -N \sin 30^\circ = -m_A a_A \cos 30^\circ \quad \textcircled{2} \end{cases}$$

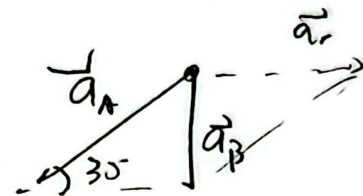
$$\textcircled{B} \quad \begin{cases} \Sigma F_y = m_B g - N_B = m_B a_B \quad \textcircled{3} \\ \Sigma F_x = 0 = 0, \quad a_B = a_A \sin 30^\circ \end{cases}$$

$$\vec{a}_B = \vec{a}_A + \vec{a}_r \quad \vec{a}_r = \text{rel}$$

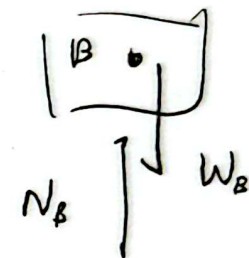
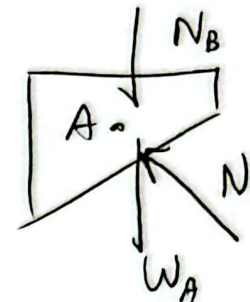
$\therefore$ 4 vars, 4 Eqs.

Solvable

$$\begin{cases} \text{a)} & a_A = 6.24 \text{ m/s}^2 \\ \text{b)} & a_r = 5.4 \text{ m/s}^2 \end{cases}$$



$$a_B = a_A \sin 30^\circ \quad \textcircled{4}$$



Q13.38 $x_{max}, F_m = ?$

(Known) $m = 5 \text{ kg}$

a) $k = 3 \text{ kN/m}$

$$F = kx, \quad U = \int F dx = \frac{1}{2} kx^2$$

$$mgx_m = \frac{1}{2} kx_m^2$$

$$x_m = \frac{2mg}{k} = 32.7 \text{ mm}$$

$$F_m = kx_m = 98.1 \text{ N}$$

b) $k' = 3(x + 160x^3) \text{ kN/m}$

$$U_p = U_g$$

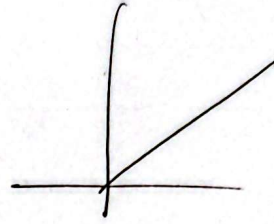
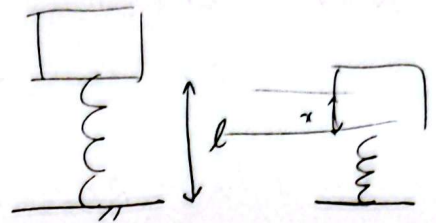
$$\int_0^{x_m'} 3(x + 160x^3) dx = mgx_m'$$

$$3\left(\frac{x_m'^2}{2} + 40x_m'^4\right) = mgx_m'$$

$$120(x_m')^3 + 0(x_m')^2 + \frac{3}{2}(x_m') - mg = 0$$

$$x_m' = 0.736 \text{ m}$$

$$F_m' = 193.578 \text{ N}$$



Q1341

$$\left\{ \begin{array}{l} W \cos 40^\circ - N_B = \frac{mv_B^2}{R} \\ 0 + mg \Delta h_{AB} = \frac{1}{2} m v_B^2 + 0 \end{array} \right.$$

$$\Delta h_{AB} = R(1 - \cos 40^\circ), R = 90 \text{ ft}$$

$$\therefore N = 167.016$$

